1.

List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

Ans1

IPv4	**IPv6**
Uses **32-bit** addresses.	Uses **128-bit** addresses.
Address looks like **192.168.1.1**.	Address looks like **2001:db8::1**.
Supports about **4.3 billion** addresses.	Supports a **huge number of addresses** (about 3.4×10^{38}).
Written in **decimal numbers** separated by dots (.).	Written in **hexadecimal** separated by colons (:).
Uses **ARP** to find the MAC address of a device.	Uses **Neighbor Discovery Protocol (NDP)**.
NAT is commonly used because IPv4 addresses are limited.	NAT is generally **not required** because of the large address space.
Broadcast communication is supported.	**No broadcast**; it uses multicast and anycast instead.
Header is more complex for some functions.	Header is **simpler and more efficient**.

2.

Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

Ans2

Network: **192.168.10.0/24**
Subnet Mask: **255.255.255.0**
Default Gateway: **192.168.10.1**

Device	IP Address	Subnet Mask	Default Gateway
💻 Laptop	192.168.10.10	255.255.255.0	192.168.10.1
📱 Phone	192.168.10.11	255.255.255.0	192.168.10.1
📺 Smart TV	192.168.10.12	255.255.255.0	192.168.10.1
🖨️ Printer	192.168.10.13	255.255.255.0	192.168.10.1

3.

Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

Ans3

/27 Subnetting for 192.168.1.0

A **/27** subnet mask is **255.255.255.224**.

- Total addresses per subnet = **32**
- Usable hosts per subnet = **32 − 2 = 30**
- From a **/24** network to **/27**, borrowed bits = 3
- Number of subnets = $\mathbf{2^3 = 8}$

First 3 Subnets

Subnet	Network Address	Usable Host IPs	Broadcast Address
1	**192.168.1.0**	192.168.1.1 – 192.168.1.30	**192.168.1.31**
2	**192.168.1.32**	192.168.1.33 – 192.168.1.62	**192.168.1.63**
3	**192.168.1.64**	192.168.1.65 – 192.168.1.94	**192.168.1.95**

4.

the subnet mask to CIDR notation and use bitwise AND for network address calculation You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet.

Hint: Convert.

ans4

Given

- **IP address:** 10.0.5.17
- **Subnet mask:** 255.255.255.0
- **CIDR notation: /24**

1. Network Address using Bitwise AND

IP: 10.0.5.17

Mask: 255.255.255.0

AND: 10.0.5.0

So, the **Network Address = 10.0.5.0**

2. Broadcast Address

For a /24 network, the last 8 bits are available for hosts.

Therefore:

Broadcast Address = 10.0.5.255

3. Valid Host Address Range

The network address and broadcast address cannot be assigned to hosts.

Valid host IPs: 10.0.5.1 – 10.0.5.254

5.

Use an online subnet calculator tool (such as https://www.subnet-calculator.com/) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

Ans5

- IP: **192.168.0.1**
- Subnet Mask: **255.255.255.0**
- Network: **192.168.0.0**
- Broadcast: **192.168.0.255**

IP Address: 10.0.5.17
Subnet Mask: 255.255.255.0

Then the result should show:

- **Network:** 10.0.5.0
- **Broadcast:** 10.0.5.255
- **Host Range:** 10.0.5.1 – 10.0.5.254

Take a new screenshot after entering those values.